

Sourav Sarkar

Full Stack Developer · AI / LLM Systems

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EXPERIENCE

Carmaa Car Care — Full Stack Developer

Sep 2025 – Present

- Promoted from intern to full-time. Owned AI agents, the multi-tenant SaaS platform, and end-to-end production infrastructure — built in TypeScript.*
- Implemented **Meta WhatsApp coexistence**, letting an owner run the WhatsApp app and the AI's automated replies on the same number answering manually or letting the agent handle it, with no change to how they already work.
 - Built an end-to-end AI sales agent on WhatsApp with **LangChain + Gemini** automating the full booking funnel (intent detection, slot booking, payments) at a **<3s** median response, over a multi-provider LLM layer with failover, rate-limit handling, and live Slack human handover.
 - Architected a multi-tenant social-media AI SaaS on **Docker Swarm** where a shared core and code-driven per-tenant containers coexist on one cluster, each business isolated with its own resource tier, config, and data, provisioned with zero manual setup.
 - Set up and ran the entire production stack on AWS from scratch hardened EC2 (ARM/Graviton) nodes, a **Docker Swarm cluster**, **ECR**, Elastic IPs, and monitoring plus rollouts and Android releases via the Google Play Console.
 - Configured **nginx** as the edge reverse proxy and **load balancer**, routing each tenant by host/path from one entry point, terminating TLS, and enabling zero-downtime rollouts.
 - Built a **Razorpay** subscription and billing subsystem end to end: plan catalog, tiered feature-gating middleware, secure payment verification, and per-account renewal/expiry reminders driving recurring revenue.
 - Delivered a full-stack SaaS dashboard with role-based access, live WebSocket updates, and per-conversation AI controls, giving owners real-time control over every customer interaction.
 - Drove growth with a lead-retargeting engine and a referral / token-wallet system, lifting booking conversion by **20%** and user acquisition by **30%**.
 - Integrated the **Meta APIs** (WhatsApp Business, Facebook, Instagram) for cross-platform messaging and automated social posting across every tenant.
 - Shipped a multimodal car-damage feature: an LLM reads vehicle images, returns structured defect descriptions, and recommends the right repair services.
 - Designed event-driven workflows on **AWS EventBridge** (Scheduler + Rules) for LinkedIn scheduling, billing reminders, and alerts; automated vehicle-info extraction with the **Google Vision API**.

Carmaa Car Care — Full Stack Developer Intern

May 2025 – Aug 2025

- Designed and restructured MongoDB schemas with backward compatibility, so old and new data worked seamlessly after schema changes; drove data migration and validation, preventing data loss.

OPEN SOURCE

PeerTube — Decentralized Video Platform (3 merged PRs)

github.com/Chocobozzz/PeerTube

- Added email alerts for logins from unrecognized devices, fingerprinting each device as a hashed record in a dedicated table and enriching alerts with GeoIP location and parsed user-agent details over the existing **Redis**-backed job queue. (PR #7737)
- Fixed the player control bar never fading out while paused, replacing a **userActive(false)** call that retriggered video.js's activity loop with direct DOM listeners so only real input resets the timer. (PR #7725)
- Built selectable **30/90/365-day and all-time channel analytics** with validated API ranges, adaptive daily/weekly/monthly aggregation, synchronized totals, a responsive chart UI, and test coverage. (PR #7750)

Velonus — Multi-Tool Security Scanner

github.com/AliAmmar15/Velonus · PR #5

- Fixed duplicate cross-tool findings with a second-pass dedup grouping by (**file**, **line**, **CWE**), keeping the highest severity.

PROJECTS

Git Gandalf — AI-Powered Git Commit Security Validator

Node.js · Ollama

- Built an AI pre-commit hook using a local LLM (Ollama) that analyzes staged changes and approves or blocks commits against configurable rules secret detection for API keys/tokens/passwords, branch protection, safe-files bypass, and a **--force** override.

SKILLS

AI / LLM	LangChain, LangGraph, Ollama, Cloud LLM APIs (Gemini), tool/agent orchestration, Google Vision API
LANGUAGES	TypeScript, JavaScript, Python
FRONTEND	React, React Native, Vite, Redux Toolkit, Tailwind CSS, Bootstrap, Framer Motion, WebSockets / Socket.io
BACKEND	Node.js, Express.js, FastAPI, REST APIs, JWT auth, LangChain agent orchestration
DATABASES	MongoDB, PostgreSQL, DynamoDB, Redis, Firebase schema design & data migration
DEVOPS / CLOUD	AWS (EC2, ECR, EventBridge), Docker, Docker Swarm, nginx, Git, CI/CD
INTEGRATIONS	Meta APIs (WhatsApp / Facebook / Instagram), Razorpay, Slack API, Google Play Console

EDUCATION

B.Tech, Mechanical Engineering — National Institute of Technology, Durgapur

Aug 2023